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Generating order recommendations for network site configurations

Abstract

A computerized method analyzes proposed orders for network sites and generates associated order recommendations. A system configuration order (SCO) manager receives a proposed order from an ordering platform and generates a proposed site configuration using the proposed order and a current site configuration of the site. In some examples, the current site configuration is obtained from a network configuration manager associated with the site. The SCO manager generates a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model. A component difference between the recommended site configuration and the proposed site configuration is identified and an order recommendation is generated using the identified component difference. The generated order recommendation is then provided to the ordering platform. The SCO manager catches ordering errors and reduces ordering inefficiencies in complex site network management systems.

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Background/Summary

BACKGROUND

(1) In large, distributed systems that include many different network sites, the management of equipment and/or component inventories throughout the network presents significant challenges. Manual placement of orders for components for the various network sites can often result in errors and/or inefficient ordering of components due to the use of incorrect equipment stock keeping units (SKUs), ordering of components that are already in inventory elsewhere in the site network, or the like. Such inefficiencies in orders throughout such a system cause significant negative impacts on the budgets of associated companies.

SUMMARY
(2) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.
(3) A computerized method for analyzing proposed orders for network sites and generating associated order recommendations is described. A system configuration order (SCO) manager receives a proposed order from an ordering platform and generates a proposed site configuration using the proposed order and a current site configuration of the site. In some examples, the current site configuration is obtained from a network configuration manager associated with the site. The SCO manager generates a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model. A component difference between the recommended site configuration and the proposed site configuration is identified and an order recommendation is generated using the identified component difference. The generated order recommendation is then provided to the ordering platform.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present description will be better understood from the following detailed description read considering the accompanying drawings, wherein:
(2) FIG. **1** is a block diagram illustrating an example system configured for generating an order recommendation using a configuration recommendation model;
(3) FIG. **2** is a sequence diagram illustrating an example process for configuring and using a system configuration order manager to provide order recommendations;
(4) FIG. **3** is a flowchart illustrating an example method for providing an order recommendation in response to a proposed order using a system configuration order manager;
(5) FIG. **4** is a flowchart illustrating an example method for training a configuration recommendation model using site configurations and performance data as training data;
(6) FIG. **5** is a diagram illustrating an example graphical user interface (GUI) for displaying an order recommendation to an ordering user; and
(7) FIG. **6** illustrates an example computing apparatus as a functional block diagram.
(8) Corresponding reference characters indicate corresponding parts throughout the drawings. In FIGS. **1** to **6**, the systems are illustrated as schematic drawings. The drawings may not be to scale. Any of the figures may be combined into a single example or embodiment.
DETAILED DESCRIPTION
(9) Aspects of the disclosure provide a system and method for analyzing proposed orders for network sites using a configuration recommendation model and generating order recommendations for those proposed orders. The disclosure is configured to receive proposed orders from an ordering platform and generate proposed site configurations using current site configurations and the proposed orders. The proposed site configurations are provided to the configuration recommendation model as input and the model generates recommended site configurations. In some examples, the configuration recommendation model is trained to generate recommended site configurations using a training set of site configurations that has been filtered based on performance data. The disclosure identifies component differences between the recommended site configurations and the proposed site configurations and generates order recommendations based on the identified component differences. The order recommendations are then provided to the ordering platform in response to the proposed orders. Thus, the disclosure uses data collected from network

configuration managers of the site network to detect errors or inefficiencies in proposed orders and to make recommendations for improving the efficiency and/or correcting errors of the proposed orders.

(10) The disclosure operates in an unconventional manner at least by integrating the network configuration managers of a distributed site network with the ordering platform thereof using the described SCO manager and associated configuration recommendation model. The disclosure uses the network configuration data and associated inventory information of the site network to analyze proposed orders and detect errors and/or inefficiencies in those orders. Further, the disclosure makes recommendations to ordering users for how to improve proposed orders, ensuring that detected errors and/or inefficiencies are reviewed and considered prior to proposed orders being completed. Thus, the disclosure reduces the monetary costs associated with the placement of erroneous or inefficient orders throughout the associated system. Additionally, the resource costs (e.g., use of processing resources and/or network resources within the system) associated with correcting for erroneous or inefficient orders after they have been placed are significantly reduced.

(11) Further, in some examples, the disclosure operates to automatically recommend orders for components that will improve and/or optimize site networks in the system, thereby reducing the time during which site networks are operating with non-optimal site configurations. Such improvements in the efficiencies of site configurations across all sites in the site network yield improved system resource use efficiency and reduction of associated resource costs for the general operations of those sites that are improved and/or optimized.

(12) FIG. **1** is a block diagram illustrating an example system **100** configured for generating an order recommendation **134** using a configuration recommendation model **120**. The system **100** includes a system configuration order (SCO) manager **102**, a network configuration manager **104**, and an ordering platform **106**. The SCO manager **102** interacts with both the network configuration manager **104** and the ordering platform **106** to improve the efficiency and accuracy of orders being placed by ordering users **108** on the ordering platform **106**. In some cases, users **108** place orders on the ordering platform **106** that include errors, incompatible components, and/or component combinations that are likely to result in inefficient site operations. By leveraging the site configuration information that is maintained and managed by the network configuration manager(s) **104** of the system **100**, the SCO manager **102** is configured to detect likely errors or other inefficiencies in proposed orders **138** before those orders are placed and to provide order recommendations **134** to the ordering users **108**, enabling the users **108** to correct any issues with the proposed orders **138**.

(13) Further, in some examples, the system **100** includes computing devices (e.g., the computing apparatus of FIG. **6**) that are configured to communicate with each other via one or more communication networks (e.g., an intranet, the Internet, a cellular network, other wireless network, other wired network, or the like). In some examples, entities of the system **100** are configured to be distributed between the multiple computing devices and to communicate with each other via network connections. For example, the SCO manager **102** includes one or more computing devices that communicate with the network configuration manager **104** and/or the ordering platform **106** over a network. Additionally, or alternatively, the network sites **110-112** include a plurality of computing devices that are in communication with each other over one or more communication networks. In other examples, other organizations of computing devices are used to implement the system **100** without departing from the description.

(14) The network configuration manager **104** includes hardware, firmware, and/or software configured for collecting, storing, and/or otherwise managing site configurations **118** associated with multiple network sites **110-112**. In some examples, the network configuration manager **104** is configured to request site configurations **114-116** from the network sites **110-112** and to receive site configurations **114-116** in response to those requests. For instance, in an example, the network configuration manager **104** periodically sends requests to each network site **110-112** with which it

is associated to maintain up-to-date site configurations **118**.
(15) Further, the network configuration manager **104** is configured to communicate and/or otherwise interact with the SCO manager **102**. In some examples, the network configuration manager **104** provides site configurations **118** to the SCO manager **102** for use as site configurations **124** in training data **122** and/or for use as current site configurations **128** to use as input with the configuration recommendation model **120**. In some such examples, the network configuration manager **104** responds to requests for site configurations **118** of specific network sites **110-112** from the SCO manager **102** for use as current site configurations **128**. Alternatively, or additionally, in some examples, the network configuration manager **104** provides the SCO manager **102** with batches of site configurations **118** periodically and/or based on a defined trigger (e.g., a quantity of site configurations **118** that have changed since the last batch was sent to the SCO manager **102** reaches a defined threshold).
(16) Additionally, or alternatively, the network configuration manager **104** generates and/or obtains performance data **126** associated with the performance of network sites **110-112** in association with the site configurations **114-116** of those network sites **110-112**. The performance data **126** is provided to the SCO manager **102** for use as part of the training data **122** used to train the configuration recommendation model **120**. In other examples, some or all of the performance data **126** used in the training data **122** is obtained by the SCO manager **102** from other sources without departing from the description.
(17) The network sites **110-112** include sites and/or other locations that include components used to maintain and/or operate a network, such as a cellular network. Such network sites **110-112** include from small pole-based distributed micro sites to macro sites that have cellular network towers used to propagate wireless signals of the cellular network and/or otherwise communicate with devices that are connected to the cellular network. Additionally, or alternatively, other types of sites are in the network sites **110-112**, such as storage sites that store inventories of components for use by other sites in the network. In other examples, other types of sites or locations are used and/or managed as described herein without departing from the description.
(18) The site configurations (e.g., site configurations **114-116**, **118**, **124**, etc.) include sets of components that are present, installed, and/or in use at specific network sites **110-112**. In some examples, the components of a site configuration **118** include hardware, firmware, and/or software components that are in use at the associated network site **110-112**. Such components can include computing devices, software and/or firmware installed on the computing devices, cables, hazard protection equipment, concealing accessories and/or other hardware used at the network site, or the like, irrespective of size.
(19) The SCO manager **102** includes hardware, firmware, and/or software configured to train a configuration recommendation model to generate order recommendations **134** based on a set of proposed components **130** and a current site configuration **128**. Further, the SCO manager **102** is configured to receive proposed components **130** of a proposed order **138** and to generate and provide an order recommendation **134** to the ordering platform **106** in response to those proposed components **130**.
(20) In some examples, the SCO manager **102** uses the training data **122** to train the configuration recommendation model **120** using machine learning techniques. For instance, in an example, the configuration recommendation model **120** includes a model trained to classify an input site configuration as being closest to one of a set of site configuration categories that are present in the site configurations **124** of the training data **122**. In such an example, site configurations **124** of the training data **122** are divided into a plurality of categories and the configuration recommendation model **120** is then trained using the site configurations **124** of the categories to classify input site configurations (e.g., the proposed site configuration **132**) as one or more of the categories. In some such examples, the performance data **126** is used to filter the site configurations **124** used during training to only include the site configurations **124** that meet or exceed a defined level of

performance (e.g., only sites that have performance data **126** indicative of performance metrics in the top 30% of site configurations are used to train the configuration recommendation model **120**).
(21) Alternatively, or additionally, the configuration recommendation model **120** includes a model trained to generate site configurations using given sets of components (e.g., proposed components **130**). The site configurations **124** are used to train the configuration recommendation model **120** by identifying site configurations **124** that have desirable performance metrics based on the associated performance data **126** and then analyzing the sets of components in those identified site configurations **124**. The configuration recommendation model **120** is trained to associate the components of an identified site configuration **124** with each other to enable the model **120** to generate predicted or recommended components for input site configurations that are incomplete, inefficient, unreliable with high break rates, or that include incompatibilities or other issues. In some examples, the components of a site configuration are identified by the associated stock keeping unit (SKU) or other identifiers and the identifiers of the components are tokenized and encoded in the model **120**. These encodings are then used to generate a set of components that are associated with both the encoded input components and also any previous components that have been generated for the output. An output set of components is compared to the proposed site configuration **132** that was provided as input and differences in the sets of components are used to generate the order recommendation **134** as described herein.
(22) Further, the SCO manager **102** provides an interface via which proposed components **130** of a proposed order **138** are obtained from an ordering platform **106**. The SCO manager **102** also obtains an identifier for the network site with which the proposed order **138** is associated. Using the obtained identifier, the current site configuration **128** of that network site is obtained and/or accessed. The current site configuration **128** and proposed components **130** are combined to form a proposed site configuration **132** and that is provided as input the configuration recommendation model **120**.
(23) In some examples, the configuration recommendation model **120** is configured to generate a recommended site configuration **133** based on the proposed site configuration **132**. The recommended site configuration **133** is generated to be similar to site configuration patterns of one or more of the site configurations **124** in the training data **122** and to be similar to the proposed site configuration **132**. Differences between the proposed site configuration **132** and the recommended site configuration **133** are used to generate the order recommendation **134**. For instance, in an example, the recommended site configuration **133** includes a component A that is not present in the proposed site configuration and lacks a component B that is part of the proposed components **130** of the proposed site configuration **132**. The SCO manager **102** identifies the difference and includes a recommendation to replace component B with component A in the order recommendation **134**.
(24) Further, in some examples, the configuration recommendation model **120** is configured to generate a recommended site configuration **133** such that components that are part of the current site configuration **128** are weighed more heavily than components in the proposed components **130**. This means that the recommended site configuration **133** is more likely to include components that are already present at the network site in the current site configuration **128** than components that have yet to be ordered and delivered to the network site. Thus, the recommended site configuration **133** is generated to provide recommendations for how to improve the current site configuration **128** rather than recommending a site configuration that requires substantial reconfiguration of the current site configuration **128**. However, it should be understood that, in some examples, the configuration recommendation model **120** generates a recommended site configuration **133** that does include component changes to the current site configuration **128** without departing from the description.
(25) Additionally, or alternatively, the SCO manager **102** has access to inventory data associated with components that are stored at network sites **110-112** and the inventory data includes components that are stored at sites while not being in use currently. In such examples, the

configuration recommendation model **120** is configured to use the inventory data when generating the recommended site configuration **133**, such that the configuration recommendation model **120** is more likely to recommend a site configuration **133** that includes components that can be obtained from other network sites **110-112** rather than components that would be ordered from outside the network. Thus, the SCO manager **102** is configured to enable the network to efficiently use components that have already been purchased and to reduce the occurrences of purchasing components that could be sourced from within the set of network sites **110-112**.

(26) The ordering platform **106** includes hardware, firmware, and/or software configured to enable a user **108** to propose and place orders (e.g., proposed order **138**) for components intended for use at one or more of the network sites **110-112**. In some examples, the ordering platform **106** includes an ordering interface **136** (e.g., a GUI interface) that enables a user **108** to select components to be included in an order, to propose an order, to update an order, and/or to confirm or complete an order. Ordering users **108** access the ordering platform **106** and generate proposed orders **138** of proposed components **130** using the ordering interface **136**.

(27) Further, the ordering platform **106** is configured to receive order recommendations **134** from the SCO manager **102** in response to providing proposed components **130** of a proposed order **138** to the SCO manager **102** for analysis as described herein. In some examples, an order recommendation **134** is displayed or otherwise provided to an ordering user **108** via the ordering interface **136** that includes one or more components that are recommended to be included in the proposed order. The ordering user **108** is enabled to update the proposed order **138** to include one or more of the recommended components or to reject the recommended components and make no changes to the proposed order **138**. Additionally, or alternatively, the order recommendation **134** includes a recommendation that one or more components be removed from the proposed order **138** (e.g., the components to be removed are incompatible with the site configuration or they can be source from within the network rather than ordered). The recommendation to remove the one or more components is displayed or otherwise provided to the ordering user **108** via the ordering interface **136** and the ordering user **108** is enabled to update the proposed order **138** by removing one or more of the components in the order recommendation **134**. In other examples, more or different recommendations of components to be added to and/or removed from the proposed order **138** are provided to the ordering user **108** via the ordering interface **136** without departing from the description.

(28) Additionally, or alternatively, the ordering interface **136** displays or otherwise provides the order recommendation **134** in the form of one or more recommended site configurations (e.g., recommended site configuration **133**). The components of a recommended site configuration are displayed to the ordering user **108** and the recommended changes between the proposed order **138** and the components of the recommended site configuration are highlighted to the ordering user **108**. Further, the user **108** is enabled to make changes to the proposed order **138** through interactions with components of the displayed recommended site configuration. Additional information about the recommended site configurations is displayed to the user, such as predicted or estimated performance metrics of the recommended site configuration (e.g., based on the performance data **126**). Thus, the ordering user **108** is provided more contextual information about the recommended site configuration and why that site configuration might be an improvement over the set of components in the proposed order **138**.

(29) In some examples, the ordering interface **136** provides or displays prompts with the order recommendation **134** to provide the ordering user **108** with more information about the recommendation **134**. For instance, in an example, the order recommendation **134** includes a recommendation to swap a component A for component B in the proposed order **138**. The SCO manager **102** detects that the component A in the proposed components **130** is actually incompatible with another part of the site configuration, while component B is compatible with the site configuration. The ordering interface **136** provides the ordering user **108** with a prompt

explaining the incompatibility, enabling the ordering user **108** quickly determine whether to accept the recommended change to the proposed order **138**. In many cases, the ordering user **108** included component A instead of component B by mistake and SCO manager **102** accurately detected the mistake. In other cases, the ordering user **108** has ordered component A intentionally for another purpose, such that the ordering user **108** does not replace component A with component B in the proposed order **138** or the ordering user **108** adds component B to the proposed order **138** while keeping component A on the proposed order **138**.

(30) Additionally, or alternatively, the SCO manager **102** is configured to automatically make changes to proposed orders based on the generated order recommendations. For instance, in some examples, if a proposed order results in a proposed site configuration that is significantly different from any known, recommended site configuration, the SCO manager **102** is configured to provide the order recommendation in the form of a changed order. In some such examples, an ordering user that prefers to proceed with the original proposed order is required to override the SCO manager **102** recommendation and/or to provide reason for proceeding with the abnormal proposed order. Such a reason may be reviewed by another user of the system and/or other parts of the system, such that ordering mistakes that result in abnormal orders are more likely to be caught within the described process.

(31) Further, in some examples, the SCO manager **102** is configured to generate proposed orders for sites without receipt of a proposed order from an ordering user. For instance, in an example, the SCO manager **102** receives a current site configuration **128** of a site and analyzes it with the configuration recommendation model **120**. The configuration recommendation model **120** generates a recommended site configuration **133** that differs significantly from the current site configuration **128**. The SCO manager **102** then generates a proposed order based on component differences between the recommended site configuration **133** and the current site configuration **128**. The proposed order is provided to the ordering platform **106** and/or to another entity in the system for review and/or completion of the order. Thus, the SCO manager **104** is configured to automatically perform order generation to maintain the site configurations of the sites in the associated site network. Further, in some examples, the SCO manager **102** is configured to generate proposed orders for parts that have been historically determined to reach a critical performance level requiring their replacement before a catastrophic fault occurs.

(32) FIG. **2** is a sequence diagram illustrating an example process **200** for configuring and using a SCO manager **102** to provide order recommendations (e.g., order recommendations **134**). In some examples, the process **200** is executed or otherwise performed in a system such as system **100** of FIG. **1**.

(33) At **202**, the SCO manager **102** registers with the network configuration manager **104**. In some examples, the registration process includes providing the network configuration manager **104** with instructions and/or associated information that enables the network configuration manager **104** to provide the SCO manager **102** with the appropriate information, such as site configuration data (e.g., current site configurations **128** and/or site configurations **124** for use in training data **122**). Further, the registration with the network configuration manager **104** regarding when and/or how to send such information, such as establishing a schedule for sending site configuration information periodically or rules for when the network configuration manager **104** sends new or updated site configuration information.

(34) At **204**, the network configuration manager **104** updates a trigger for when and/or how to send information to the SCO manager **102** based on the registration of the SCO manager **102** at **202**. The updated trigger includes software that detects one or more trigger states and based on that detection, causes the network configuration manager **104** to send site configuration information of one or more network sites **110-112** to the SCO manager **102**. For instance, in an example, the network configuration manager **104** is triggered to send site configuration information of a network site **110** to the SCO manager **102** upon detecting that the site configuration of the network site **110**

has been updated or otherwise changed.

(35) At **206**, the network configuration manager **104** requests a configuration from a network site **110**. In some examples, the network configuration manager **104** requests configurations from a plurality of different network sites **110-112**. Such requests are sent to network sites periodically or according to some other pattern. Additionally, or alternatively, the network configuration manager **104** requests configurations from network sites **110-112** based on a triggering event. Further, in some examples, the network site **110** is configured to send a notification and/or its configuration to the network configuration manager **104** upon the configuration being changed.

(36) At **208**, the network site **110** determines its current configuration and sends the configuration to the network configuration manager **104** in response to the request at **206**. In some examples, the network site **110** is configured to perform software operations to determine the set of components that currently make up the configuration of the network site **110**.

(37) At **210**, the network configuration manager **104** sends the configuration of the network site **110** to the SCO manager **102**. In some examples, the network configuration manager **104** is configured to send the configuration to the SCO manager **102** based on receiving the configuration from the network site **110**. Alternatively, or additionally, the network configuration manager **104** is configured to send configurations received from network sites to the SCO manager **102** periodically and/or based on the occurrence of a triggering event (e.g., the quantity of site configurations to be sent to the SCO manager **102** reaches a defined threshold).

(38) At **212**, the SCO manager **102** stores the received configuration. In some examples, the stored configuration is used as part of training data (e.g., training data **122**) for use in training the configuration recommendation model **120** of the SCO manager **102**. Alternatively, or additionally, the stored configuration is used as a current site configuration (e.g., current site configuration **128**) when the SCO manager **102** is analyzing a proposed order (e.g., proposed order **138**) and providing an order recommendation (e.g., order recommendation **134**) associated therewith.

(39) At **214**, an ordering user **108** enters a proposed order into the ordering platform **106**. It should be understood that, in some examples, the entering of the proposed order at **214** occurs some time after the storing of the configuration at **212** and, in other examples, the process from **202-212** is performed during substantially the same time period as the process from **214-224** without departing from the description.

(40) At **216**, the ordering platform **106** requests an order recommendation from the SCO manager **102**. In some examples, the request for the order recommendation includes providing the SCO manager **102** with data from the proposed order, such as the proposed components (e.g., proposed components **130**). Further, the request for the order recommendation includes an identifier or indicator of the network site with which the proposed order is associated, enabling the SCO manager **102** to access the current site configuration of that network site for use in generating the order recommendation as described herein.

(41) At **218**, the SCO manager **102** generates and sends the recommendation (e.g., the order recommendation **134**) to the ordering platform **106** and, at **220**, the recommendation is presented to the ordering user **108** by the ordering platform **106** (e.g., via an ordering interface **136**). In some examples, the generation of the recommendation includes generating one or more recommended site configurations and populating the recommendation with component changes required to create one of the recommended site configurations at the associated network site, as described herein.

(42) At **222**, the ordering user **108** updates the proposed order using the presented recommendation. In some examples, the ordering user **108** is enabled to remove components, add components, swap components, and/or make other changes to the proposed order through interaction with the presented recommendation. For instance, in an example, the presented recommendation includes a list of the components of a recommended site configuration and highlighted portions indicating the changes that need to be made to the proposed order to achieve the recommended site configuration. In such an example, the ordering user **108** is enabled to select the recommended site configuration

and, through that selection, cause the proposed order to be changed according to the highlighted portions of the presented recommendation. In other examples, the ordering user **108** is enabled to update the proposed order using other methods without departing from the description.

(43) At **224**, after updating the proposed order, the ordering platform **224** completes the order, such that the order is placed or otherwise initiated. In some examples, as a result of the presented recommendation from the SCO manager **102**, the ordering user **108** is enabled to avoid errors and/or inefficiencies in the proposed order.

(44) FIG. **3** is a flowchart illustrating an example method **300** for providing an order recommendation (e.g., order recommendation **134**) in response to a proposed order (e.g., proposed order **138**) using a SCO manager (e.g., SCO manager **102**). In some examples, the method **300** is executed or otherwise performed in a system such as system **100** of FIG. **1**.

(45) At **302**, a proposed order is received from an ordering platform. In some examples, the proposed order includes a set or group of proposed components that are to be ordered and/or other configuration information associated with the proposed components of the proposed order.

(46) At **304**, a proposed site configuration is generated using the proposed order and a current site configuration of the site for which the proposed order is being proposed. In some examples, the current site configuration is stored by and accessed by the SCO manager **102**. Alternatively, in some examples, the current site configuration is requested from a network configuration manager (e.g., network configuration manager **104**) with which the site is associated.

(47) Further, in some examples, generating the proposed site configuration includes combining components of the site that are present in the current site configuration with the proposed components of the proposed order. In some such examples, the SCO manager **102** is configured to determine when a proposed component is likely to be added to the components of the current site configuration and/or when a proposed component is likely to be used to replace an existing component of the current site configuration. When such a determination is made, the proposed site configuration is generated to include an indication of whether a proposed component is intended to replace an existing component or just be added to the set of existing components. Such determinations are made using specification data of components that provides information about how components interact with each other, capacities for components to be connected with other components, and/or requirements for components that a particular component needs to operate.

(48) At **306**, a recommended site configuration (e.g., recommended site configuration **133**) is generated using a configuration recommendation model (e.g., the configuration recommendation model **120**). In some examples, the proposed site configuration, including data of the current site configuration and the proposed order, are used as input to the configuration recommendation model and the configuration recommendation model classifies the proposed site configuration as being similar to the recommended site configuration and/or the configuration recommendation model otherwise generates the recommended site configuration. For instance, in an example, the configuration recommendation model is trained to generate components of the recommended site configuration based on the components present in the proposed site configuration and based on components that have already been generated for the recommended site configuration. Further, in some examples, the recommended site configuration is generated based on inventory data of the network of sites of which the site is a part, such that components of the recommended site configuration are more likely to be components that can be obtained from elsewhere in the site network rather than purchased from outside the site network.

(49) At **308**, a component difference between the recommended site configuration and the proposed site configuration is identified. In some examples, the component difference includes a component that is present in the recommended site configuration but not present in the proposed site configuration. Alternatively, or additionally, the component difference includes a component that is present in the proposed site configuration but not present in the recommended site configuration. In other examples, other types of component differences are identified, such as differing versions of

components, without departing from the description.
(50) At **310**, an order recommendation is generated using the identified component difference. In some examples, the order recommendation includes a recommendation to include a component in the order that is present in the recommended site configuration and/or to remove a component from the order that is not present in the recommended site configuration. In some such examples, the order recommendation includes a recommendation to replace a first component with a second component in the order, wherein the first component is present in the proposed site configuration and the second component is present in the recommended site configuration.
(51) Further, in some examples, the SCO manager uses the configuration recommendation model to generate a plurality of recommended site configurations and then identifies a plurality of component differences between each recommended site configuration and the proposed site configuration. In some such examples, the generated order recommendation includes one or more recommendations associated with each recommended site configuration of the plurality recommended site configurations. Further, the order recommendation is generated to include information that describes or otherwise provides context information for each of the recommended site configurations, such that an ordering user that views the order recommendation can make an informed decision regarding which recommended site configuration to emulate, if any.
(52) At **312**, the generated order recommendation is provided to the ordering platform. In some examples, the order recommendation is then displayed or otherwise presented to an ordering user via an ordering interface (e.g., ordering interface **136**). Further, in some examples, the ordering user using the ordering platform is enabled to make changes to the proposed order based on the order recommendation using the ordering interface.
(53) FIG. **4** is a flowchart illustrating an example method **400** for training a configuration recommendation model (e.g., configuration recommendation model **120**) using site configurations (e.g., site configurations **124**) and performance data (e.g., performance data **126**) as training data (e.g., training data **122**). In some examples, the method **400** is executed or otherwise performed in a system such as system **100** of FIG. **1**.
(54) At **402**, a plurality of site configurations and associated performance data are obtained from a site network. In some examples, the site configurations and performance data are obtained from one or more network configuration managers (e.g., network configuration manager **104**).
(55) At **404**, a site configuration of the plurality of site configurations is selected. At **406**, if the performance data of the selected site configuration does not meet a defined threshold or thresholds, the selected site configuration is filtered out of the plurality of site configurations at **408**. Alternatively, if the performance data of the selected site configuration does meet the defined threshold or thresholds, the process proceeds to **410**.
(56) At **410**, if site configurations remain to be selected in the plurality of site configurations, the process returns to **404** to select another site configuration that has not yet been selected. Alternatively, if no site configurations remain to be selected the process proceeds to **412**.
(57) At **412**, the configuration recommendation model is trained to categorize input site configurations as site configurations of the filtered plurality of site configurations. Alternatively, or additionally, the configuration recommendation model is trained to generate groups or sets of components in the form of recommended site configurations based on patterns of components that are present in the filtered plurality of site configurations as described herein.
(58) FIG. **5** is a diagram illustrating an example graphical user interface (GUI) **500** for displaying an order recommendation (e.g., order recommendation **134**) to an ordering user (e.g., ordering user **108**). In some examples, the GUI **500** is displayed or otherwise presented to a user as part of an ordering interface, such as ordering interface **136** of the ordering platform **106** in system **100** of FIG. **1**.
(59) The GUI **500** includes a proposed order portion **502** that displays information about the current proposed order. As illustrated, the displayed information includes a list of components that

are being ordered for the site and an identifier of the site itself. It should be understood that, in other examples, more and/or different information about the proposed order is displayed without departing from the description.

(60) The GUI **500** further includes an order recommendation portion **504**. As illustrated, the order recommendation portion **504** displays recommendations and information associated with multiple recommended site configurations, including a recommended config. #1 and a recommended config. #2. The order recommendation portion **504** displays contextual information about each of the recommended site configurations, including a performance metric at which the first recommended site configuration excels and a quantity of other sites that use the second recommended site configuration. In other examples, more, less, and/or different context information is provided about recommended site configurations without departing from the description.

(61) Additionally, the order recommendation portion **504** as illustrated provides order change buttons **506**, **508**, and **510** to enable an ordering user to quickly and easily make changes to the proposed order that are in line with one of the recommended site configurations. For instance, in an example, a user selects the order change button **506** and the proposed order is changed by replacing the order for component B with an order for component D. Alternatively, in another example, the user selects the order change buttons **508** and **510** and the proposed order is changed adding an order for a component E and an order for two of component F. In other examples, more, fewer, and/or different recommended changes are displayed in the portion **504** without departing from the description. Further, in other examples, more and/or different types of user interface components are used without departing from the description.

(62) The GUI **500** includes a save order button **512** and a complete order button **514**. The save order button **512** enables the ordering user to save the proposed order in its current state for later access and the complete order button **514** causes the proposed order to be sent out for completion.

(63) Exemplary Operating Environment

(64) The present disclosure is operable with a computing apparatus according to an embodiment as a functional block diagram **600** in FIG. **6**. In an example, components of a computing apparatus **618** are implemented as a part of an electronic device according to one or more embodiments described in this specification. The computing apparatus **618** comprises one or more processors **619** which may be microprocessors, controllers, or any other suitable type of processors for processing computer executable instructions to control the operation of the electronic device. Alternatively, or in addition, the processor **619** is any technology capable of executing logic or instructions, such as a hard-coded machine. In some examples, platform software comprising an operating system **620** or any other suitable platform software is provided on the apparatus **618** to enable application software **621** to be executed on the device. In some examples, analyzing proposed orders to generate order recommendations as described herein is accomplished by software, hardware, and/or firmware.

(65) In some examples, computer executable instructions are provided using any computer-readable media that is accessible by the computing apparatus **618**. Computer-readable media include, for example, computer storage media such as a memory **622** and communications media. Computer storage media, such as a memory **622**, include volatile and non-volatile, removable, and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or the like. Computer storage media include, but are not limited to, Random Access Memory (RAM), Read-Only Memory (ROM), Erasable Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), persistent memory, phase change memory, flash memory or other memory technology, Compact Disk Read-Only Memory (CD-ROM), digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage, shingled disk storage or other magnetic storage devices, or any other non-transmission medium that can be used to store information for access by a computing apparatus. In contrast,

communication media may embody computer readable instructions, data structures, program modules, or the like in a modulated data signal, such as a carrier wave, or other transport mechanism. As defined herein, computer storage media does not include communication media. Therefore, a computer storage medium should not be interpreted to be a propagating signal per se. Propagated signals per se are not examples of computer storage media. Although the computer storage medium (the memory **622**) is shown within the computing apparatus **618**, it will be appreciated by a person skilled in the art, that, in some examples, the storage is distributed or located remotely and accessed via a network or other communication link (e.g., using a communication interface **623**).

(66) Further, in some examples, the computing apparatus **618** comprises an input/output controller **624** configured to output information to one or more output devices **625**, for example a display or a speaker, which are separate from or integral to the electronic device. Additionally, or alternatively, the input/output controller **624** is configured to receive and process an input from one or more input devices **626**, for example, a keyboard, a microphone, or a touchpad. In one example, the output device **625** also acts as the input device. An example of such a device is a touch sensitive display. The input/output controller **624** may also output data to devices other than the output device, e.g., a locally connected printing device. In some examples, a user provides input to the input device(s) **626** and/or receives output from the output device(s) **625**.

(67) The functionality described herein can be performed, at least in part, by one or more hardware logic components. According to an embodiment, the computing apparatus **618** is configured by the program code when executed by the processor **619** to execute the embodiments of the operations and functionality described. Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (ASICs), Program-specific Standard Products (ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), Graphics Processing Units (GPUs).

(68) At least a portion of the functionality of the various elements in the figures may be performed by other elements in the figures, or an entity (e.g., processor, web service, server, application program, computing device, or the like) not shown in the figures.

(69) Although described in connection with an exemplary computing system environment, examples of the disclosure are capable of implementation with numerous other general purpose or special purpose computing system environments, configurations, or devices.

(70) Examples of well-known computing systems, environments, and/or configurations that are suitable for use with aspects of the disclosure include, but are not limited to, mobile or portable computing devices (e.g., smartphones), personal computers, server computers, hand-held (e.g., tablet) or laptop devices, multiprocessor systems, gaming consoles or controllers, microprocessor-based systems, set top boxes, programmable consumer electronics, mobile telephones, mobile computing and/or communication devices in wearable or accessory form factors (e.g., watches, glasses, headsets, or earphones), network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like. In general, the disclosure is operable with any device with processing capability such that it can execute instructions such as those described herein. Such systems or devices accept input from the user in any way, including from input devices such as a keyboard or pointing device, via gesture input, proximity input (such as by hovering), and/or via voice input.

(71) Examples of the disclosure may be described in the general context of computer-executable instructions, such as program modules, executed by one or more computers or other devices in software, firmware, hardware, or a combination thereof. The computer-executable instructions may be organized into one or more computer-executable components or modules. Generally, program modules include, but are not limited to, routines, programs, objects, components, and data

structures that perform particular tasks or implement particular abstract data types. Aspects of the disclosure may be implemented with any number and organization of such components or modules. For example, aspects of the disclosure are not limited to the specific computer-executable instructions, or the specific components or modules illustrated in the figures and described herein. Other examples of the disclosure include different computer-executable instructions or components having more or less functionality than illustrated and described herein.

(72) In examples involving a general-purpose computer, aspects of the disclosure transform the general-purpose computer into a special-purpose computing device when configured to execute the instructions described herein.

(73) An example system comprises: a processor; and a memory comprising computer program code, the memory and the computer program code configured to cause the processor to: receive a proposed order associated with a site from an ordering platform; generate a proposed site configuration using the proposed order and a current site configuration of the site; generate a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identify a component difference between the recommended site configuration and the proposed site configuration; generate an order recommendation using the identified component difference; and provide the generated order recommendation to the ordering platform.

(74) An example computerized method comprises: receiving a proposed order associated with a site from an ordering platform; generating a proposed site configuration using the proposed order and a current site configuration of the site; generating a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identifying a component difference between the recommended site configuration and the proposed site configuration; generating an order recommendation using the identified component difference; and providing the generated order recommendation to the ordering platform.

(75) One or more computer storage media having computer-executable instructions that, upon execution by a processor, cause the processor to at least: receive a proposed order associated with a site from an ordering platform; generate a proposed site configuration using the proposed order and a current site configuration of the site; generate a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identify a component difference between the recommended site configuration and the proposed site configuration; generate an order recommendation using the identified component difference; and provide the generated order recommendation to the ordering platform.

(76) Alternatively, or in addition to the other examples described herein, examples include any combination of the following: wherein identifying the component difference between the recommended site configuration and the proposed site configuration includes identifying one or more of the following: a component in the recommended site configuration that is not present in the proposed site configuration; and a component in the proposed site configuration that is not present in the recommended site configuration. wherein identifying the component difference between the recommended site configuration and the proposed site configuration includes: identifying a first component in the recommended site configuration that is not present in the proposed site configuration; identifying a second component in a proposed order that is not present in the recommended site configuration; and wherein the generated order recommendation includes a recommendation to replace the second component with the first component in the proposed order. wherein generating the order recommendation uses inventory data of a site network associated with the site and the inventory data indicates that the second component is available in the site network. further comprising training the configuration recommendation model, the training including: obtaining a plurality of site configurations and associated performance data from a site network

associated with the site; filtering site configurations that are associated with performance data below a defined performance threshold from the plurality of site configurations; and training the configuration recommendation model to receive an input site configuration and to identify a site configuration of the filtered plurality of site configurations that is similar to the input site configuration. wherein generating a recommended site configuration includes generating a plurality of recommended site configurations; wherein identifying the component difference includes identifying component differences between the proposed site configuration and each recommended site configuration of the plurality of recommended site configurations; and wherein the generated order recommendation includes a plurality of recommendations for changes to the proposed order associated with each recommended site configuration of the plurality of recommended site configurations. further comprising: determining an identifier of the site; requesting the current site configuration of the site from a network configuration manager using the determined identifier; and receiving the requested current site configuration from the network configuration manager.
(77) Any range or device value given herein may be extended or altered without losing the effect sought, as will be apparent to the skilled person.
(78) Examples have been described with reference to data monitored and/or collected from the users (e.g., user identity data with respect to profiles). In some examples, notice is provided to the users of the collection of the data (e.g., via a dialog box or preference setting) and users are given the opportunity to give or deny consent for the monitoring and/or collection. The consent takes the form of opt-in consent or opt-out consent.
(79) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.
(80) It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages. It will further be understood that reference to 'an' item refers to one or more of those items.
(81) The embodiments illustrated and described herein as well as embodiments not specifically described herein but within the scope of aspects of the claims constitute an exemplary means for receiving a proposed order associated with a site from an ordering platform; exemplary means for generating a proposed site configuration using the proposed order and a current site configuration of the site; exemplary means for generating a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; exemplary means for identifying a component difference between the recommended site configuration and the proposed site configuration; exemplary means for generating an order recommendation using the identified component difference; and exemplary means for providing the generated order recommendation to the ordering platform.
(82) The term "comprising" is used in this specification to mean including the feature(s) or act(s) followed thereafter, without excluding the presence of one or more additional features or acts.
(83) In some examples, the operations illustrated in the figures are implemented as software instructions encoded on a computer readable medium, in hardware programmed or designed to perform the operations, or both. For example, aspects of the disclosure are implemented as a system on a chip or other circuitry including a plurality of interconnected, electrically conductive elements.
(84) The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and examples of the disclosure may include additional or fewer operations than those disclosed herein. For example, it is contemplated

that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure.
(85) When introducing elements of aspects of the disclosure or the examples thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The term “exemplary” is intended to mean “an example of” The phrase “one or more of the following: A, B, and C” means “at least one of A and/or at least one of B and/or at least one of C.”
(86) Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined in the appended claims. As various changes could be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A system comprising: a processor; and a memory comprising computer program code, the memory and the computer program code configured to cause the processor to: obtain a plurality of site configurations associated with a plurality of network sites based on an occurrence of a triggering event; receive a proposed order associated with a network site of the plurality of network sites from an ordering platform, wherein the proposed order comprises a set of proposed components to be ordered together; generate a proposed site configuration by combining the set of proposed components of the proposed order and a set of existing components of a current site configuration of the network site; generate a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identify a component difference between the recommended site configuration and the proposed site configuration; generate an order recommendation using the identified component difference, wherein the order recommendation comprises a plurality of recommended changes; provide the generated order recommendation to the ordering platform; present, via an order interface of the ordering platform, a site identifier of the network site, the set of proposed components of the proposed order, and the plurality of recommended changes; receive, via the order interface, an ordering user's interaction with the presented plurality of recommended changes; cause the proposed order to be changed according to a recommended change of the plurality of recommended changes based on the ordering user's interaction; obtain performance data associated with the plurality of site configurations; compare the performance data to a defined performance threshold; filter the plurality of site configurations to remove site configuration that are associated with performance data below the defined performance threshold; and generate a training set of site configurations comprising the filtered plurality of site configurations, wherein the configuration recommendation model is further trained using the training set of site configurations.
2. The system of claim 1, wherein the plurality of recommended changes comprise a recommendation to remove an incompatible component from the proposed order.
3. The system of claim 1, wherein the triggering event comprises one or more of the plurality of network sites sending a notification upon a configuration change.
4. The system of claim 1, wherein each component of the filtered plurality of site configurations is identified by a corresponding component identifier of a plurality of component identifiers, and wherein the plurality of component identifiers are tokenized and encoded in the configuration recommendation model.
5. The system of claim 1, wherein the memory and the computer program code are configured to

further cause the processor to: present, via the ordering interface, a set of order change buttons, each of the set of order change buttons associated with a corresponding recommended change of the plurality of recommended changes.

6. The system of claim 1, wherein generating the recommended site configuration includes generating a plurality of recommended site configurations; wherein identifying the component difference includes identifying component differences between the proposed site configuration and each recommended site configuration of the plurality of recommended site configurations; wherein the plurality of recommended changes includes a plurality of recommendations for changes to the proposed order associated with each recommended site configuration of the plurality of recommended site configurations; wherein each of the plurality of recommended changes is displayed with contextual information; and wherein the contextual information comprises a quantity of other sites of the plurality of network sites that use a specific site configuration of the plurality of recommended site configuration.

7. The system of claim 1, wherein the plurality of network sites comprise small pole-based distributed micro sites and cellular network tower-based macro sites.

8. A computerized method comprising: obtaining a plurality of site configurations associated with a plurality of network sites based on an occurrence of a triggering event; receiving a proposed order associated with a network site of the plurality of network sites from an ordering platform, wherein the proposed order comprises a set of proposed components to be ordered together; generating a proposed site configuration by combining the set of proposed components of the proposed order and a set of existing components of a current site configuration of the network site; generating a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identifying a component difference between the recommended site configuration and the proposed site configuration; generating an order recommendation using the identified component difference, wherein the order recommendation comprises a plurality of recommended changes; providing the generated order recommendation to the ordering platform; presenting, via an order interface of the ordering platform, a site identifier of the network site, the set of proposed components of the proposed order, and the plurality of recommended changes; receiving, via the order interface, an ordering user's interaction with the presented plurality of recommended changes; causing the proposed order to be changed according to a recommended change of the plurality of recommended changes based on the ordering user's interaction; obtaining performance data associated with the plurality of site configurations; comparing the performance data to a defined performance threshold; filtering the plurality of site configurations to remove site configuration that are associated with performance data below the defined performance threshold; and generating a training set of site configurations comprising the filtered plurality of site configurations, wherein the configuration recommendation model is further trained using the training set of site configurations.

9. The computerized method of claim 8, wherein the plurality of recommended changes comprise a recommendation to remove an incompatible component from the proposed order.

10. The computerized method of claim 8, wherein the triggering event comprises one or more of the plurality of network sites sending a notification upon a configuration change.

11. The computerized method of claim 8, wherein each component of the filtered plurality of site configurations is identified by a corresponding component identifier of a plurality of component identifiers, and wherein the plurality of component identifiers are tokenized and encoded in the configuration recommendation model.

12. The computerized method of claim 8, further comprising: presenting, via the ordering interface, a set of order change buttons, each of the set of order change buttons associated with a corresponding recommended change of the plurality of recommended changes.

13. The computerized method of claim 8, wherein generating a recommended site configuration includes generating a plurality of recommended site configurations; wherein identifying the

component difference includes identifying component differences between the proposed site configuration and each recommended site configuration of the plurality of recommended site configurations; and wherein the plurality of recommended changes includes a plurality of recommendations for changes to the proposed order associated with each recommended site configuration of the plurality of recommended site configurations; wherein each of the plurality of recommended changes is displayed with contextual information; and wherein the contextual information comprises a quantity of other sites of the plurality of network sites that use a specific site configuration of the plurality of recommended site configuration.
14. The computerized method of claim 8, wherein plurality of network sites comprise small pole-based distributed micro sites and cellular network tower-based macro sites.
15. A computer storage medium has computer-executable instructions that, upon execution by a processor, cause the processor to at least: obtain a plurality of site configurations associated with a plurality of network sites based on an occurrence of a triggering event; receive a proposed order associated with a network site of the plurality of network sites from an ordering platform, wherein the proposed order comprises a set of proposed components to be ordered together; generate a proposed site configuration by combining the set of proposed components of the proposed order and a set of existing components of a current site configuration of the network site; generate a recommended site configuration using a configuration recommendation model and the proposed site configuration as input to the configuration recommendation model; identify a component difference between the recommended site configuration and the proposed site configuration; generate an order recommendation using the identified component difference, wherein the order recommendation comprises a plurality of recommended changes; provide the generated order recommendation to the ordering platform; present, via an order interface of the ordering platform, a site identifier of the network site, the set of proposed components of the proposed order, and the plurality of recommended changes; receive, via the order interface, an ordering user's interaction with the presented plurality of recommended changes; cause the proposed order to be changed according to a recommended change of the plurality of recommended changes based on the ordering user's interaction; obtain performance data associated with the plurality of site configurations; compare the performance data to a defined performance threshold; filter the plurality of site configurations to remove site configuration that are associated with performance data below the defined performance threshold; and generate a training set of site configurations comprising the filtered plurality of site configurations, wherein the configuration recommendation model is further trained using the training set of site configurations.
16. The computer storage medium of claim 15, wherein the plurality of recommended changes comprise a recommendation to remove an incompatible component from the proposed order.
17. The computer storage medium of claim 15, wherein the triggering event comprises one or more of the plurality of network sites sending a notification upon a configuration change.
18. The computer storage medium of claim 15, wherein each component of the filtered plurality of site configurations is identified by a corresponding component identifier of a plurality of component identifiers, and wherein the plurality of component identifiers are tokenized and encoded in the configuration recommendation model.
19. The computer storage medium of claim 15, wherein the computer-executable instructions that, upon execution by a processor, further cause the processor to: present, via the ordering interface, a set of order change buttons, each of the set of order change buttons associated with a corresponding recommended change of the plurality of recommended changes.
20. The computer storage medium of claim 15, wherein generating the recommended site configuration includes generating a plurality of recommended site configurations; wherein identifying the component difference includes identifying component differences between the proposed site configuration and each recommended site configuration of the plurality of recommended site configurations; wherein the plurality of recommended changes includes a

plurality of recommendations for changes to the proposed order associated with each recommended site configuration of the plurality of recommended site configurations; wherein each of the plurality of recommended changes is displayed with contextual information; and wherein the contextual information comprises a quantity of other sites of the plurality of network sites that use a specific site configuration of the plurality of recommended site configuration.